13/08/24 Module 2 Linear Algebra 2

1. Vector space
2. Subspace
3. Linear dependence & Independence
4. Spanning of subspace
5. Basis & dimension.
6. Inner product
7. Gram Schmidt orthogonal process
8. Linear transformation

~~Vector space~~

1 A binary operation $*$ on a non-empty set S is a mapping of $S \times S$ into S or $*: S \times S \to S$ (from, cross)

Eg: + is a binary operator on $Z \times Z$

2 Algebraic Structure

A non-empty set equipped with 1 or more binary operators is called an algebraic structure.

Eg: $(N,+)$. $(R,+,\cdot)$, $(P(A), \cup)$ (crypt P / bold P)

3 Group

The algebraic structure $G\,*$ is a group if it satisfies

G_1: Closure property $a*b = c \in G \ \forall a,b \in G$

G_2: Associativity property $a*(b*c) = (a*b)*c$, $\forall a,b,c \in G$

G_3: Extreme of identity $\Rightarrow a*e = e*a = a$, ~~$\forall a \in G$~~ $\forall a \in G$

G_4: $\forall a \in G$, $\exists$ (there exist) $a^{-1} \in G$ $\Rightarrow a*a^{-1} = a^{-1}*a = e$, identity in G

eg $(Z,+)$ is a group. integers follow the properties for addⁿ

4 Abelian group

Group $(G,+)$ is abelian if it satisfies

$G_5 = a+b = b+a \ \forall a,b \in G$, commutativ

Eg: $(Z,+)$ is a abelian group

5 Order of a group

It is the number of elements in the group

6 Sub group

A non-empty subset H of a group G is a subgroup iff ① $\forall a,b \in H$, $a*b \in H$ (closure) iff → if & only if
② $\forall a \in H$, $a^{-1} \in H$ (inverse)

Eg: $Z = \{\dots,-3,-2,-1,0,1,2,3,\dots\}$ is a group w.r.to +

Set of even integers $E = \{\dots,-6,-4,-2,0,2,4,6\dots\}$ is a subgroup of Z w.r.to +.

7 Field

A non empty set with 2 binary operators $*$ & $\cdot$ is a field if it satisfies

1. $(F,*)$ is an abelian grp
2. $(F^*,\cdot)$ is an abelian grp, where $F^* = F - \{0\}$
3. $a\cdot(b*c) = a\cdot b * a\cdot c$
   $\forall a,b,c \in F$

Eg: $(R,+,\cdot)$ is a Field

8. Vector Space

Let F be a field and V be a non empty set on which a binary operator + is defined. V is said to be a vector space over the field F denoted by V(F).

If 1. V is an abelian group w.r.to +
2. There is another operator called scalar multiplication, such that (S)

for every $\forall \alpha \in F$ and $v \in V$, $\alpha v \in V$

satisfying the following

1. $\alpha(u+v) = \alpha u + \alpha v$, $\alpha \in F$, $u, v \in V$
2. $(\alpha+\beta)v = \alpha v + \alpha\beta$, $\alpha, \beta \in R$, $v \in V$
3. $(\alpha\beta)v = \alpha(\beta v) \in V$, $\alpha, \beta \in F$, $v \in V$
4. $1v = v \in V$

Elements of V are called vectors and elements of F are called scalars.

eg. $R(C)$, R is not a vector space over C.

Justification, $\alpha = (3+2i) \in C = F$.

$v = 5 \in R = V$

$\alpha v = (15+10i) \notin R = V$

9. Subspace

Let V(F) be a non empty subset W of V is W itself ie W(F) and if

① $x, y \in W$, $x+y \in W$

② $\alpha \in F$, $x \in W \Rightarrow \alpha x \in W$

The single axiom combining ① & ②

W is subspace iff for $\alpha, \beta \in F$, $x, y \in W$, $\alpha x + \beta y \in W$

eg.1. Set of all 2x2 symmetric matrices ~~from~~ form a subspace of all 2x2 matrices over R.

Let $A = \begin{bmatrix} a & c \\ c & b \end{bmatrix}$, $B = \begin{bmatrix} d & f \\ f & e \end{bmatrix}$

$$\alpha A + \beta B = \begin{bmatrix} \alpha a & \alpha c \\ \alpha c & \alpha b \end{bmatrix} + \begin{bmatrix} \beta d & \beta f \\ \beta f & \beta e \end{bmatrix} = \begin{bmatrix} \alpha a+\beta d & \alpha c+\beta f \\ \alpha c+\beta f & \alpha b+\beta e \end{bmatrix}$$

which is a 2x2 symmetric matrix

$\therefore \alpha A + \beta B \in W$

eg2. Show that the set of all 2x2 non singular matrix is not a subspace of set of all 2x2 matrices over R

$\alpha = \beta = 1 \in R$, $1 \in R$, $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $|A| \neq 0$ non singular

$B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ $|B| \neq 0$ "

$\alpha A + \beta B = 1\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 1\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, singular

10. Linear Combination

Let V(F), let $\{v_1, v_2, \dots, v_n\}$ be a set of n vectors at V

Let $x \in V$ can be written as $x = c_1v_1 + c_2v_2 + \dots + c_nv_n$, where $c_1, c_2 \dots c_n \in F$

Then x is linear combination of $v_1, v_2, \dots, v_n$

11. Linear Span

Let V(F) and $S \subset V$ and $S = \{v_1, v_2, \dots, v_n\}$

The set of all linear combination of S is known as linear span of S, L(S).

$L(S)$ is a subspace of $V(F)$

$L(S)$ is smallest subspace of V containing S

$L(S)$ is subspace of V generated by S

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12. Linear dependent vectors

Let $V(F)$, $v_1, v_2, \cdots v_n \in V$ and

$\alpha_1, \alpha_2, \cdots, \alpha_n \in F$

$v_1, v_2, \ldots v_n$ are called linearly dependent

if $\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = 0$

$\Rightarrow$ not all α_i's are zeroes

13. Linear independent vectors

$\alpha_1 v_1 + \alpha_2 v_2 + \quad + \alpha_n v_n = 0 \Rightarrow \alpha_1 = \alpha_2 = \cdots = = \alpha_n = 0$

14. Basis and Dimension

Let $V(F)$, the subset $\{v_1, v_2, \ldots, v_n\}$ of V is a basis of V if $v_1, v_2, \ldots v_n$ are linearly independent and $\{v_1, v_2, \ldots v_n\}$ span V. Dimension $\rightarrow$ no of elements in the basis eg $R^3 \rightarrow 3$ dimension

eg $e_1 = (1,0,0), e_2 = (0,1,0), e_3 = (0,0,1)$

$\{e_1, e_2, e_3\}$ forms a basis for R^3

Let $(a,b,c) \in R^3$, $(a,b,c) = ae_1 + be_2 + ce_3$

$\alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3 = 0 \Rightarrow (\alpha_1, \alpha_2, \alpha_3) = (0,0,0)$

$\Rightarrow \alpha_1 = \alpha_2 = \alpha_3 = 0$ linearly independent

15. Inner Product

An inner product is generalisation of dot product

1. $\langle x, y \rangle = \langle y, x \rangle$
2. $\langle x, x \rangle \geq 0$
3. $\langle x+y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$
4. $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$

Inner product space

A vector space together with an inner product

eg: $e_1 = (a,b), e_2 = (c,d)$

$\langle e_1, e_2 \rangle = ac + bd$

Norm of (x,y,z)

$\|(x,y,z)\| = \sqrt{x^2+y^2+z^2}$

Normalization of (x,y,z) is $\left(\frac{x}{\|(x,y,z)\|}, \frac{y}{\|(x,y,z)\|}, \frac{z}{\|(x,y,z)\|}\right)$

Let R^3 be an inner product space with std inner product

1. Find Norm of $(3,0,4)$

Ans: $\|(3,0,4)\| = \sqrt{9+0+16} = \sqrt{25} = \underline{\underline{5}}$

2. Normalize $(1/2, -1/3, 1/4)$

$= (6/12, -4/12, 3/12)$ $\quad \|(6/12, -4/12, 3/12)\| = \sqrt{\frac{36+16+9}{144}} = \underline{\underline{\frac{\sqrt{61}}{12}}}$

Normalization vector is $\left(\frac{6}{\sqrt{61}}, \frac{-4}{\sqrt{61}}, \frac{3}{\sqrt{61}}\right)$

16. Gram-Schmidt Orthogonalization

Let $V(F)$, and $\{v_1, v_2, \ldots, v_n\}$ be a basis of V.

To find orthonormal basis, we shall construct basic vectors as follows.

$$w_1 = \frac{v_1}{\|v_1\|}, \quad v_1 \neq 0$$

$$w_2 = \frac{u_2}{\|u_2\|} \text{ where } u_2 = v_2 - \langle v_2, w_1 \rangle w_1$$

$$w_3 = \frac{u_3}{\|u_3\|} \text{ where } u_3 = v_3 - \langle v_3, w_1 \rangle w_1 - \langle v_3, w_2 \rangle w_2$$

17. Linear Transformation

Let V, W be two vector spaces over F. A mapping $T: V \to W$ is said to be a linear Transformation, If it satisfies the following:

1. $T(v_1+v_2) = T(v_1)+T(v_2)$, $\forall v_1, v_2 \in V$
2. $T(\alpha v_1) = \alpha T(v_1)$, $\forall v_1 \in V, \alpha \in F$

Image set of Linear Transformation

$\text{Im } T = \{T(v) / v \in V\}$

$\ker T$ kernel T, $\ker T = \{v / T(v) = 0\}$

Exercise 1

1. Show that $(z, +)$ is an abelian group

Ans: $z = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$

$\forall a, b \in z$, then $a+b \in z$

$(z,+)$ satisfies closure property.

$\forall a, b, c \in z$, then $a+(b+c) = (a+b)+c$

$(z,+)$ satisfies associativity

clearly $0 \in z$ and $\forall a \in z, a+0 = 0+a = a$

$\exists$ identity 0 in $(z,+)$

$\forall a \in z, \exists -a \in z$ s.t. $a+(-a) = (-a)+a = 0$

ie, $\forall a \in z$, $\exists$ inverse element

$+$ is commutative

$\forall a, b \in z, a+b = b+a$ $\therefore (z,+)$ is an abelian group

Q. Show that $(z-\{0\}, \cdot)$ is not a group

Ans: $10 \in z - \{0\}$

$e = 1 \in z - \{0\}$ $\forall a \in z, \exists -a \in z$

$\nexists$ inverse for 10

HW

Q. S.T $(Q-\{0\}, \cdot)$ is an abelian group

Ans: $Q - \{0\} \Rightarrow$

$\forall a, b \in Q - \{0\}, a \cdot b \in Q - \{0\}$

$\forall a, b, c \in Q - \{0\}, a\cdot(b \cdot c) = (a \cdot b)\cdot c$

$\forall a \in Q - \{0\}, a \cdot 1/a = 1$

ie $1/a$ is inverse element

$\forall a, b \in Q - \{0\} \Rightarrow a \cdot b = b \cdot a$

$\therefore Q - \{0\}$ is a abelian grp

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1Q. S.T Set of all non-singular square matrices whose entries are rational, real or complex numbers under the operation matrix multiplication is a group but not an abelian grp.

Ans: Let M be the set of all $n \times n$ square matrices,

Let $A, B \in M$. Then AB is a square matrix of order n

$\therefore A, B \in M \Rightarrow M$ is closed under matrix multiplication

As per definition of matrix multiplication $A(BC) = (AB)C$

$\therefore$ Matrix multiplication is associative

Let I_n be the identity matrix

$$AI_n = I_nA = A$$

$\therefore I_n \in M$ is the identity element.

$A^{-1} = \frac{1}{|A|} adj(A) \in M \quad \therefore A^{-1}$ is inverse of $A \in M$

As per definition of matrix multiplication $AB \neq BA$

2Q. Let $G = \{1, -1, i, -i\}$ under multiplication of complex nos

S.T $(G, \cdot)$ is an abelian grp.

·	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

From the table

1. $(G, \cdot)$ satisfies closure property
2. Multiplication of complex no.s is associative.
3. $\exists\ e = 1$, the identity in G
4. Inverse of $1 = 1$

   $-1 = -1$

   $i = \frac{1}{i} = \frac{i}{i^2} = -i$

   $-i = i$
5. Multiplication of complex nos is commutative

$\therefore (G, \cdot)$ is an abelian grp.

3Q. S.T $H = \{1, -1\}$ is a sub grp of $(G, \cdot)$ when $G = \{1, -1, i, -i\}$

Ans: ① $H \subset G$

② $\forall a, b \in H$

$1 \cdot -1 = -1 \in H$

$1 \cdot 1 = 1 \in H$

$-1 \cdot -1 = 1 \in H$

③ $\forall a \in H, a^{-1} \in H \qquad \therefore (H, \cdot)$ is a sub grp of $(G, \cdot)$

$1^{-1} = 1 \in H$

$(-1)^{-1} = -1 \in H$

4Q. Show that $R^2 = \{(a,b) / a, b \in R\}$, the set of all ordered pair of real numbers is an abelian grp w.r.to the operation ~~com~~ coordinate wise addition.

Ans: $(a,b) + (c,d) = (a+c, b+d)$

G_1: Closure property

$\forall (a_1, b_1), (a_2, b_2) \in R^2$

Then $(a_1, b_1) + (a_2, b_2) = (a_1 + a_2, b_1 + b_2) \in R^2$

G_2: $\forall (a_1, b_1), (a_2, b_2), (a_3, b_3) \in R^2$

$$(a_1, b_1) + [(a_2, b_2) + (a_3, b_3)] = (a_1, b_1) + [a_2 + a_3, b_2 + b_3]$$
$$= (a_1 + a_2 + a_3, b_1 + b_2 + b_3) - ①$$

Now $[(a_1, b_1) + (a_2, b_2)] + (a_3, b_3) = (a_1 + a_2 + a_3, b_1, b_2, b_3) - ②$

From ① & ②

$(a_1, b_1) + [(a_2, b_2) + (a_3, b_3)] = [(a_1, b_1) + (a_2, b_2)] + (a_3, b_3)$

R^2 satisfies associativity

G_3: $\forall (a,b) \in R^2, \exists (0,0) \in R^2$ s.t. $(a,b)+(0,0) = (a+0, b+0)$
$= (a,b)$
$(0,0)$ is identity in R^2 $= (0,0)+(a,b)$

G_4: $\forall (a,b) \in R^2 \; \exists (-a,-b) \in R^2$ s.t. $(a,b)+(-a,-b) = (0,0)$
$\therefore \exists$ inverse for every $(a,b) \in R^2$

G_5: $\forall (a,b),(c,d) \in R^2$, $(a,b)+(c,d) = (a+c, b+d)$
$= (c+a, d+b)$
$= (c,d)+(a,b)$

$\therefore R^2$ satisfies commutative

5 Q S.T $F[x]$, the set of all polynomials in x over a field F is a vector space w.r.to addition of polynomials $(f+g)(x) = f(x)+g(x)$, $\forall f(x), g(x) \in F(x)$ and scalar multiplication $(\alpha f)(x) = \alpha f(x)$ $\forall \alpha \in F, f(x) \in F(x)$.

Ans: Sum of 2 polynomials is a polynomial
G_1 $\therefore [F(x),+]$ is closed under $+$

G_2: $[f(x)+g(x)] + r(x) = f(x) + [g(x)+r(x)]$, $\forall f(x), g(x), r(x) \in F[x]$
Since $+$ is addition of polynomials
$\therefore (F[x],+)$ is associative

G_3: $\forall f(x) \in F[x], \exists 0 \in F[x]$ s.t. $0 + f(x) = f(x)+0 = f(x)$
$\therefore 0 \in F[x]$ is the identity

G_4 $\forall f(x) \in F(x), \exists -f(x) \in F[x]$ s.t. $f(x) + -f(x) = 0$
$\therefore \exists$ inverse for all $f(x) \in F[x]$

G_5: $\forall f(x), g(x) \in F[x]$, $f(x)+g(x) = g(x)+f(x)$
$\therefore (F[x],+)$ is an abelian grp

S_1: $\alpha\{f(x)+g(x)\} = \alpha f(x) + \alpha g(x)$, $\forall \alpha \in F$ and $f(x), g(x) \in F[x]$

S_2: $(\alpha+\beta) f(x) = \alpha f(x) + \beta f(x)$, $\forall \alpha,\beta \in F$ and $f(x) \in F[x]$

S_3: $(\alpha\beta)[f(x)] = \alpha \cdot [\beta f(x)]$ $\forall \alpha,\beta \in F$ and $f(x) \in F[x]$

S_4: $1 \in F$ s.t. $\forall f(x) \in F[x]$, $1 \cdot f(x) = f(x) \cdot 1 = f(x)$
$\therefore F[x]$ is a vector space.

6 S.T R^n is not a vector space under the equation coordinate wise addition and scalar multiplication $\alpha x = (\alpha x_1, x_2, \cdots, x_n)$, where $x = (x_1, x_2 \cdots x_n) \in R^n$

Ans Let $x, y \in R^n$ and $\alpha, \beta \in F$, when $x = (x_1, x_2, \cdots x_n)$
Based on coordinate wise addition $y = (y_1, y_2, \cdots y_n)$

$(\alpha+\beta)x = ((\alpha+\beta)x_1, x_2, \cdots x_n)$ — ①
$\alpha x = (\alpha x_1, x_2, \cdots x_n)$
$\beta x = (\beta x_1, x_2, \cdots x_n)$
$\alpha x + \beta x = (\alpha x_1 + \beta x_1, 2x_2, 2x_3, \cdots, 2x_n)$ — ②
From ① and ② $(\alpha+\beta)x \neq \alpha x + \beta x$
$\therefore R^n$ is not a vector space

7Q. S.T R^n is not a vector space under $x+y=(x_1-y_1, x_2-y_2 \cdots)$
where $x=(x_1,x_2,\cdots,x_n)$, $y=(y_1,y_2,\cdots,y_n)$ and $\cancel{\alpha x=(\alpha x_1, \alpha x_2)}$
$\alpha x=(-\alpha x_1, -\alpha x_2 \cdots)$

Ans: $x+y=(x_1-y_1, x_2-y_2,\cdots,x_n-y_n)$
$y+x=(y_1-x_1, y_2-x_2,\cdots,y_n-x_n)$

$x+y \neq y+x$ So $(R^n,+)$ is not abelian grp and so it is not a vector space

8Q. S.T R^2 is not a vector space under $x+y=(x_1+y_1, 0)$
$\alpha x=(\alpha x_1, 0)$ when $x=(x_1,x_2)$
$y=(y_1,y_2)$

Ans: $x=(1x_1,0)=(x_1,0)\neq(x_1,x_2)=x$
$1.x\neq x$ $\therefore$ $(R^2,+)$ with given scalar multiplication is not a vector space

9Q. Is the set of all polynomials of degree n a vector space under $(f+g)(x)=f(x)+g(x)$, $(\alpha f)(x)=\alpha\cdot f(x)$

Ans: Let $f(x)=a_1x^n+a_2x^{n-1}+\cdots+a_{n-1}x+a_n \in F[x]$
$g(x)=-a_1x^n$
Then $f(x)+g(x)=a_2x^{n-1}+a_3x^{n-2}+\cdots+a_{n-1}x+a_n \in F[x]$,
the polynomials of degree n
$\therefore$ Violates the closure property

10Q. S.T $R^3=L(S)$ where $S=\{e_1,e_2,e_3\}$ $e_1=(1,0,0)$, $e_2=(0,1,0)$ and $e_3=(0,0,1)$

Ans: Let $(a,b,c)\in R^3$
$(a,b,c)=ae_1+be_2+ce_3$

$\therefore$ ~~S space~~ S spans ~~space~~ $R^3 \Rightarrow R^3=L(S)$

11Q. Let $S=\{1,t,t^2,\cdots,t^n\}$ and $V=F[t]$, Set of all polynomials
S.T $L(S)=V$

Ans: Let $P(t)\in F[t]$
$P(t)=a(1)+b(t)+c(t^2)+\cdots$
$\therefore$ S spans $V=F(t)$ or $L(S)=V$
$\therefore$ Violates closure property.

12. Find the subspace of R^3 spanned by $x=(0,1,-1)$, $y=(1,0,2)$

Ans: Consider $ax+by=(0,a,-a)+(b,0,2b)$
$=(b,a,2b-a)$
$L(S)=\{(b,a,2b-a)/a,b\in R\}$

13. Show that $(1,1)$ and $(-3,2)$ are linearly independent in R^2 over R

Ans: consider $a(1,1)+b(-3,2)=(0,0)$
$(a-3b, a+2b)=(0,0)$
$a-3b=0$
$a+2b=0$
$\Rightarrow a=b=0$

14. Determine whether the vectors $(1,3,2)$, $(1,-7,8)$ and $(2,1,-1)$ are linearly independent

consider $a(1,3,2)+b(1,-7,8)+c(2,1,-1)=(0,0,0)$
$a+b+2c=0$
$3a-7b+c=0$
$2a+8b-c=0$

Short cut method

Consider $\begin{vmatrix} 1 & 1 & 2 \\ 3 & -7 & 1 \\ 2 & 8 & -1 \end{vmatrix} = 0$. $\therefore$ Rank < No. of unknown

$\therefore$ System has many solutions and $a=b=c=0$ is not a solution.

$\therefore$ Not linearly independent.

15 Express the vector $(1,-2,5)$ as linear combination of vectors $(1,1,1)$, $(1,2,3)$ and $(2,-1,1)$

Ans: $(1,-2,5) = a(1,1,1)+b(1,2,3)+c(2,-1,1)$
$= au_1 + bu_2 + cu_3$

$a+b+2c = 1$ - ①

$a+2b-c=-2$ - ②

$a+3b+c=5$ - ③

② - ① $\Rightarrow b-3c=-3$ - ④

③ - ① $\Rightarrow 2b-c=4$ - ⑤

④ $\times 2$, $2b-6c=-6$ -

$2b-c=4$

$-5c=-10$ $\therefore c=\underline{\underline{2}}$

$b=3$, $\underline{\underline{a=-6}}$

$\therefore (1,-2,5) = \underline{\underline{-6u_1+3u_2+2u_3}}$

16. Prove that $f_1=(1,2,1)$, $f_2=(2,1,0)$, $f_3=(1,-1,2)$ forms a basis for R^3.

Ans: To S.T

① f_1, f_2, f_3 are linearly independent

② $L(f_1,f_2,f_3) = R^3$

① Let $af_1+bf_2+cf_3 = (0,0,0)$

$a+2b+c=0$

$2a+b-c=0$

$a+0b+2c=0$

Consider $\begin{vmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 0 & 2 \end{vmatrix} = 2-10-1=-9 \neq 0$

$\therefore a=b=c=0$

f_1, f_2, f_3 are linearly independent.

Let $(x,y,z) \in R^3$

To S.T $(x,y,z) = af_1+bf_2+cf_3$, for some $a,b,c \in R$

$x = a+2b+c$

$y = 2a+b-c$

$z = a+2c$

$c = \frac{1}{9}[3z-2y+x]$

a = function in (x,y,z)

b = function in (x,y,z)

$\therefore (x,y,z)$ can be written as linear combination of f_1, f_2 and f_3

$\therefore L\{f_1,f_2,f_3\} = R^3$

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17. Define field. Prove that $(R,+,\cdot)$ is a field. ~~A Field $(F,+,\cdot)$ is an algebraic structure that satisfies the foll~~

Ans: A Field $(F,+,\cdot)$ is an algebraic structure that satisfies the following axioms

1. $(F,+)$ is an abelian group
2. $(F^*,\cdot)$ is an abelian group where $F^* = F - \{0\}$
3. $\forall a,b,c \in F$, $a\cdot(b+c) = a\cdot b + a\cdot c$

$(R,+,\cdot)$ is a field. Reason: 1. $(R,+)$ is an abelian group.
2. $(R-\{0\},\cdot)$ is an abelian group.
3. $\forall a,b,c \in R$, $a\cdot(b+c) = a\cdot b + a\cdot c$

18. Prove that $S = \{(1,0,0),(1,1,0),(1,1,1)\}$ forms a basis for R^3.

Ans: To prove that
1. S is linearly independent set.
2. S spans R^3, or $L(S) = R^3$

1. S is linearly independent

Let $a,b,c \in F$

$a(1,0,0)+b(1,1,0)+c(1,1,1) = (0,0,0)$

$a+b+c=0$
$b+c=0$
$c=0$

$a=b=c=0$ ∴ S is linearly independent

2. $L(S)=R^3$

Let $(x,y,z) \in R^3$

$(x,y,z) = a(1,0,0)+b(1,1,0)+c(1,1,1)$

$a+b+c=x$ $\quad c=z$
$b+c=y$ $\quad b=y-c=y-z$
$c=z$ $\quad a=x-y-z+z$

$a=x-y$ $\cancel{}$, $b=y-z$, $c=z$

19. Let V be a vector space of set of all 2×2 matrix. Show that $\dim(V)=4$

Ans: It is enough to S.T V has a basis of 4 elements

Let $S = \left\{\begin{bmatrix}1&0\\0&0\end{bmatrix},\begin{bmatrix}0&1\\0&0\end{bmatrix},\begin{bmatrix}0&0\\1&0\end{bmatrix},\begin{bmatrix}0&0\\0&1\end{bmatrix}\right\}$

To S.T, S is a basis for V or $L(S)=V$

1. S is linearly independent

consider $a\begin{bmatrix}1&0\\0&0\end{bmatrix}+b\begin{bmatrix}0&1\\0&0\end{bmatrix}+c\begin{bmatrix}0&0\\1&0\end{bmatrix}+d\begin{bmatrix}0&0\\0&1\end{bmatrix}=\begin{bmatrix}0&0\\0&0\end{bmatrix}$

$\begin{bmatrix}a&b\\c&d\end{bmatrix}=\begin{bmatrix}0&0\\0&0\end{bmatrix} \Rightarrow a=b=c=d=0$

2. $L(S)=V$

Let $\begin{bmatrix}x&y\\z&w\end{bmatrix}\in V$

$\begin{bmatrix}x&y\\z&w\end{bmatrix}=x\begin{bmatrix}1&0\\0&0\end{bmatrix}+y\begin{bmatrix}0&1\\0&0\end{bmatrix}+z\begin{bmatrix}0&0\\1&0\end{bmatrix}+w\begin{bmatrix}0&0\\0&1\end{bmatrix}$

20. Let V be a vector space of set of all 2×2 symmetric matrices. Show that $\dim(V)=3$.

Ans: It is enough to S.T V has a basis of 3 elements

Let $S=\left\{\begin{bmatrix}1&0\\0&0\end{bmatrix},\begin{bmatrix}0&1\\1&0\end{bmatrix},\begin{bmatrix}0&0\\0&1\end{bmatrix}\right\}$

1. S is linearly independent

consider $a\begin{bmatrix}1&0\\0&0\end{bmatrix}+b\begin{bmatrix}0&1\\1&0\end{bmatrix}+c\begin{bmatrix}0&0\\0&1\end{bmatrix}=\begin{bmatrix}0&0\\0&0\end{bmatrix}$

$\begin{bmatrix}a&b\\b&c\end{bmatrix}=\begin{bmatrix}0&0\\0&0\end{bmatrix}\Rightarrow a=b=c=0$

2. $L(S)=V$

Let $\begin{bmatrix}x&y\\y&w\end{bmatrix}=x\begin{bmatrix}1&0\\0&0\end{bmatrix}+y\begin{bmatrix}0&1\\1&0\end{bmatrix}+w\begin{bmatrix}0&0\\0&1\end{bmatrix}$

21. Let V be a vector space of all 2×2 matrices and

$W=\left\{\begin{bmatrix}x&-x\\y&z\end{bmatrix} \middle/ x,y,z\in R\right\}$

a) P.T W is a subspace of V
b) Find $\dim(W)$

Ans: a) Let ~~α,β~~ $\alpha,\beta\in R$, and $x_1,x_2\in W$ then to S.T

$\alpha x_1+\beta x_2\in W$

$x_1=\begin{bmatrix}x_1&-x_1\\y_1&z_1\end{bmatrix}$

$x_2=\begin{bmatrix}x_2&-x_2\\y_2&z_2\end{bmatrix}$

$\alpha x_1+\beta x_2=\begin{bmatrix}\alpha x_1+\beta x_2 & -\alpha x_1-\beta x_2\\ \alpha y_1+\beta y_2 & \alpha z_1+\beta z_2\end{bmatrix}\in W$

b. Let $S = \left\{ \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

Then S is a basis for V and so $\dim(V) = 3$

22. Let $V = R^4$ and W be a subspace of V generated by $(1,-2,5,-3)$, $(2,3,1,-4)$ and $(3,8,-3,-5)$

1. Find Basis and $\dim(W)$
2. Extend the basis of W to the basis of R^4.

Ans: 1. $A = \begin{bmatrix} 1 & -2 & 5 & -3 \\ 2 & 3 & 1 & -4 \\ 3 & 8 & -3 & -5 \end{bmatrix}$ $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$

$\sim \begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 14 & -18 & 4 \end{bmatrix}$ $R_2 \to R_2/7$

$\sim \begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 1 & -9/7 & 2/7 \\ 0 & 14 & -18 & 4 \end{bmatrix}$ $R_3 \to R_3 - 14R_2$

$\sim \begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 1 & -9/7 & 2/7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $\therefore \text{Rank}(A) = 2 = \dim(W)$

Basis of W is $\{(1,-2,5,-3), (0,1,-9/7,2/7)\}$

A can be extended to

$\sim \begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 1 & -9/7 & 2/7 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ Then A has rank $= 4 = \dim(V)$

$\therefore$ Extended Basis for V is

$\{(1,-2,5,-3), (0,1,-9/7,2/7), (0,0,1,0), (0,0,0,1)\}$

23. Show that $(1,0), (0,1)$ form an orthonormal basis of R^2

Ans: Let $e_1 = (1,0)$
$e_2 = (0,1)$

$\langle e_1, e_2 \rangle = (1\times 0) + (0 \times 1)$
$= 0 + 0 = 0$

$\Rightarrow e_1 \perp e_2$ (orthogonal)

Also $\|e_1\| = \sqrt{1^2 + 0^2} = 1$
$\|e_2\| = \sqrt{0^2 + 1^2} = 1$

$\therefore \{e_1, e_2\}$ is an orthonormal basis of R^2

24. S.T if $u = (\cos\theta, \sin\theta)$, $v = (-\sin\theta, \cos\theta)$ then $\{u, v\}$ form an orthonormal basis for R^2

Ans: Let $u = (\cos\theta, \sin\theta)$
$v = (-\sin\theta, \cos\theta)$

$\langle u, v \rangle = (\cos\theta \times -\sin\theta) + (\sin\theta \times \cos\theta)$
$= -\sin\theta\cos\theta + \sin\theta\cos\theta = 0$

$\Rightarrow u \perp v$ (orthogonal)

Also $\|u\| = \sqrt{\cos^2\theta + \sin^2\theta} = 1$
$\|v\| = \sqrt{(-\sin^2\theta) + \cos^2\theta} = 1$

$\therefore \{u, v\}$ is an orthonormal basis of R^2

25. Find an orthonormal basis for subspace W of R^3 generated by the vectors $(2,1,1)$ and $(1,3,-1)$

Ans: using Gram Schimdt Orthogonalization process.

$w_1 = \frac{v_1}{\|v_1\|}$, $v_1 = (2,1,1)$

$\|v_1\| = \sqrt{2^2+1^2+1^2} = \sqrt{6}$

$= \left(\frac{2}{\sqrt6}, \frac{1}{\sqrt6}, \frac{-1}{\sqrt6}\right)$

$u_2 = v_2 - \langle v_2, w_1 \rangle w_1$

$= (1,3,-1) - \left[\frac{2}{\sqrt6} + \frac{3}{\sqrt6} - \frac{1}{\sqrt6}\right]\left(\frac{2}{\sqrt6}, \frac{1}{\sqrt6}, \frac{+1}{\sqrt6}\right)$

$= (1,3,-1) - \left[\frac{4}{\sqrt6}\right]\left(\frac{2}{\sqrt6}, \frac{1}{\sqrt6}, \frac{1}{\sqrt6}\right)$

$= (1,3,-1) - (4/3, 2/3, 2/3) = \left(\frac{-1}{3}, 7/3, -5/3\right)$

$w_2 = \frac{u_2}{\|u_2\|} = \frac{3}{\sqrt{1+49+25}}\left(-1/3, 7/3, 5/3\right)$

$w_2 = \left(\frac{-1}{\sqrt{75}}, \frac{7}{\sqrt{75}}, \frac{-5}{\sqrt{177}}\right)$

$\therefore \{w_1, w_2\}$ forms an orthogonal basis of R^3.

26. Let $T: R^3 \to R$ be a mapping defined by $T(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^3$. ~~calculate~~ check whether T is a ~~linear~~ linear transform.

Ans: If T is a LT then $T(V_1+V_2) = T(V_1) + T(V_2)$

Let $V_1 = (x_1, x_2, x_3)$, $T(V_1) = x_1^2 + x_2^2 + x_3^2$

$V_2 = (y_1, y_2, y_3)$, $T(V_2) = y_1^2 + y_2^2 + y_3^2$

$V_1 + V_2 = (x_1+y_1, x_2+y_2, x_3+y_3)$

$T(V_1+V_2) = (x_1+y_1)^2 + (x_2+y_2)^2 + (x_3+y_3)^2 \to ①$

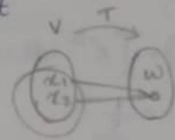
$T(V_1) + T(V_2) = x_1^2 + x_2^2 + x_3^2 + y_1^2 + y_2^2 + y_3^2 \to ②$

From ① & ② $T(V_1+V_2) \neq \cancel{T(V)}\ T(V_1) + T(V_2)$

$\therefore$ T is not a LT

27. Let V and W be two vector spaces and $T: V \to W$ be a linear transformation. Prove that

1. Ker T is a subspace of V
2. Im T is a subspace of W



Ans ① Let $V_1, V_2 \in \ker T$

$T(V_1) = 0$ and $T(V_2) = 0$

$\alpha(0) = 0$

a. Let $\alpha \in F$. Then $T(\alpha V_1) = \alpha T(V_1) = \cancel{\alpha(0)}\ 0$

b. $T(V_1 + V_2) = T(V_1) + T(V_2)$ since T is LT

$= 0 + 0 = 0$

$\therefore \alpha V_1 \in \ker T$ and and $V_1 + V_2 \in \ker T$

$\Rightarrow$ Ker T is a subspace of V.

② Let $w_1, w_2 \in \operatorname{Im} T$

$\Rightarrow \exists\, V_1, V_2 \in V \ni T(V_1) = w_1$ and $T(V_2) = w_2$

Let $\alpha \in F$, $\alpha w_1 = \alpha T(V_1) = T(\alpha V_1)$, since T is LT

$\Rightarrow \alpha w_1$ is image of αV_1 under T

$\Rightarrow \alpha w_1 \in W$.

$w_1 + w_2 = T(V_1) + T(V_2) = T(V_1 + V_2)$ since T is a LT

$\Rightarrow w_1 + w_2$ is image of $V_1 + V_2$ under T $\Rightarrow w_1 + w_2 \in W$

$\therefore$ Im T is a subspace of W

27/03/24 Open Book Exam

Module - II

1. P.T $(R, \cdot)$ is not a group.
2. Check whether $R(C)$ a vector space
3. check whether $C(R)$ a vector space
4. P.T the set of all 2x2 symmetric matrices is a subspace of set of all 2x2 matrices.
5. Show that the set of 2x2 non-singular matrices cannot be a subspace
6. check whether $(1,2,3)$ $(3,2,1)$ $(2,4,6)$ are linearly independent
7. Create an orthonormal basis with vectors $(1,1,2)$ and $(3,1,-2)$
8. Find the dim(W), where W is a subspace of R^3 generated by the vectors $(1,1,1)$, $(2,2,2)$ and $(3,3,3)$. Extend a basis of W to form a basis for R^3.
9. Let $V = \left\{ \begin{bmatrix} a & b \\ a & c \end{bmatrix} / a,b,c \in R \right\}$. Form a basis for V. Find dim(V)
10. Prove that Ker T is subspace of V and Im T a subspace of W if $T: V \to W$ a L.T where V(F), W(F) are vector spaces.
11. A map $T: R^2 \to R$ such that $T(x,y) = \frac{x+y}{2}$ check whether T is a Linear Transformation.
12. P.T $\{(1,0,0,0), (0,1,0,0) (0,0,1,0) (0,0,0,1)\}$ form an orthonormal basis of R^4.
13. Find subspace of R^4 spanned by $(1,0,2,-1)$ and $(0,1,4,-2)$

Answers

1. ~~$10 \in R$~~ $0 \in R$
Inverse of 0 does not exist
$\therefore (R, \cdot)$ is not a group

2. $R(C)$,
$\alpha = (3+2i) \in C = F$
$V = 5 \in R = V$
$\alpha V = (15+10i) \notin R = V$
$\therefore R(C)$, R is not a vector space over C

3. $C(R)$
~~a, b ∈ C, then a+~~
$a+ib, x+iy \in C$, then $a+ib+x+iy \in C$
$(C,+)$ satisfies closure property.

$\forall a+ib, x+iy, p+iq \in C$
then $a+ib+(x+iy+p+iq) = (a+ib+x+iy)+p+iq$
$(C,+)$ satisfies associativity

clearly $0 \in C$ and $\forall a+ib \in C$, $a+ib+0 = 0+a+ib = a+ib$
$\exists$ identity 0 in $(C,+)$

$\forall a+ib \in z$, $\exists -(a+ib) \in C$ $\ni$ $a+ib+(-a-ib) = -a-ib+a+ib = 0$
ie $\forall a+ib \in C$, $\exists$ inverse elements

+ is commutative
$\forall a+ib, x+iy \in C$, $a+ib+x+iy = x+iy+a+ib$
$\therefore (C,+)$ is an abelian group.

V_1, V_2 is a complex no. $\in C$.

S_1. $\alpha(V_1+V_2) = \alpha V_1 + \alpha V_2$

S_2. $(\alpha+\beta)V_1 = \alpha V_1 + \beta V_1$

S_3. $\alpha(\beta V_1) = (\alpha\beta)V_1$

S_4. $1 \cdot V_1 = V_1$

4. Let $A = \begin{bmatrix} a & c \\ c & b \end{bmatrix}$, $B = \begin{bmatrix} d & f \\ f & e \end{bmatrix}$

$$\alpha A + \beta B = \begin{bmatrix} \alpha a & \alpha c \\ \alpha c & \alpha b \end{bmatrix} + \begin{bmatrix} \beta d & \beta f \\ \beta f & \beta e \end{bmatrix}$$

$$= \begin{bmatrix} \alpha a + \beta d & \alpha c + \beta f \\ \alpha c + \beta f & \alpha b + \beta e \end{bmatrix}$$ which is a 2×2 symmetric matrix.

5. $1 \in R$ $\quad A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $|A| \neq 0$ Non-singular

$\alpha = \beta = 1$

$B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ $|B| \neq 0$ Non-singular

$$\alpha A + \beta B = 1\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 1\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$ singular

$\Rightarrow$ Not singular

$\therefore$ not a subspace

6. consider $a(1,2,3) + b(3,2,1) + c(2,4,6)$

$a + 3b + 2c = 0$

$2a + 2b + 4c = 0$

$3a + b + 6c = 0$

shortcut

$$\begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 2 & 4 & 6 \end{vmatrix} = 0$$

$\therefore$ not linearly independent.

7. using Gram Schmidt orthogonal

$\omega_1 = \frac{V_1}{\|V_1\|}$ $\quad V_1 = (1,1,2)$

$\|V_1\| = \sqrt{1^2+1^2+2^2} = \sqrt{6}$

$$\omega_1 = \left(\frac{1}{\sqrt6}, \frac{1}{\sqrt6}, \frac{2}{\sqrt6}\right)$$

$u_2 = V_2 - \langle V_2, \omega_1 \rangle \omega_1$

$$= (3,1,-2) - \left[\frac{3}{\sqrt6} + \frac{1}{\sqrt6} - \frac{4}{\sqrt6}\right]\left(\frac{1}{\sqrt6}, \frac{1}{\sqrt6}, \frac{2}{\sqrt6}\right)$$

$$\omega_2 = \frac{u_2}{\|V_2\|} = \frac{3}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{-2}{\sqrt{14}}$$

(ω_1, ω_2) form the basis of orthogonal basis of R^3

8. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$ $\quad R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$

$$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank $(A) = 1 = \dim(W)$

This matrix can be extended to rank - 3

$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then

Now A has rank $3 = \dim(\omega)$

Extended basis for V is

$\{(1,1,1), (0,1,0), (0,0,1)\}$

9 $S = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

S is a basis for V & dim = 3

10. Let $V_1, V_2 \in \ker T$

$T(V_1) = 0$ & $T(V_2) = 0$

ⓐ Let $\alpha \in F$. Then $T(\alpha V_1) = \alpha T(V_1) = \alpha(0) = 0$

ⓑ $T(V_1 + V_2) = T(V_1) + T(V_2) = 0 + 0 = 0$ [$\because T$ is L.T]

From ⓐ & ⓑ, $\alpha(V_1) \in \ker T$ & $V_1 + V_2 \in \ker T$

$\Rightarrow \ker T$ is a subspace of V.

Let $\omega_1, \omega_2 \in \text{Im}\, T$

$\Rightarrow \exists\, V_1, V_2 \in V$ such that

$T(V_1) = \omega_1$ & $T(V_2) = \omega_2$

Let $\alpha \in F$. $\alpha(\omega_1) = T(V_1) = T(\alpha V_1)$ is a L.T

$\Rightarrow \alpha\omega_1$ is an image of αV_1 and $\alpha\omega_1 \in \omega$

$\omega_1 + \omega_2 = T(V_1) + T(V_2) = T(V_1 + V_2)$

[$\because$ T is a linear transformation]

$\Rightarrow \omega_1 + \omega_2$ is an image of $V_1 + V_2$ and T.

$\Rightarrow \omega_1 + \omega_2 \in \omega$

$\therefore$ Im T is a subspace of ω

11. If T is an L.T then $T(V_1 + V_2) = T(V_1) + T(V_2)$

Let $V_1 = (x, y)$, $T(V_1) = \frac{x+y}{2}$

$V_2 = (z, t)$, $T(V_2) = \frac{z+t}{2}$

$V_1 + V_2 = (x+z, y+t)$

$T(V_1 + V_2) = \frac{x+z+y+t}{2}$ — (1)

$T(V_1) + T(V_2) = \frac{x+y+z+t}{2}$ — (2)

(1) = (2)

$\therefore$ It is a linear transformation.

12. Let $P_1 = (1,0,0,0)$

$P_2 = (0,1,0,0)$

$P_3 = (0,0,1,0)$

$P_4 = (0,0,0,1)$

$\langle P_1, P_2 \rangle = \langle P_2, P_3 \rangle = \langle P_3, P_4 \rangle$

$= \langle P_1, P_2, P_3, P_4 \rangle = 0$

$e_1 \perp e_2 \perp e_3 \perp e_4$

Also $\|e_1\| = \|e_2\| = \|e_3\| = \|e_4\| = 1$

$\therefore \{e_1, e_2, e_3, e_4\}$ is an orthogonal basis of R^4

13. Consider $ax + by$

$(a, 0, 2a, -a) + (0, b, 4b, -2b)$

$= (a, b, 2a + 4b, -2b - a)$

$L(S) = \{(a, b, 2a+4b, -a-2b) \mid a, b \in R\}$

04/04/24 Module 2

1. Show that $(C, +, \cdot)$ is a field.
2. Show that $\{(1,0,0)\ (0,1,0)\ (0,0,1)\}$ is a basis for R^3
3. Is $Z[C]$ a vector space? Justify your answer.
4. Determine whether the vectors $(1,2,3)\ (3,1,2)\ (2,4,6)$ linearly independent.
5. Let V be a vector space of all 3x3 matrices show that $\dim(V)=9$.
6. Find an orthonormal basis for the subspace W of R^3 generated by the vectors $(1,1,2)$ and $(2,1,3)$.
7. Let $T: V(F) \longrightarrow W(F)$ be a linear transformation. Prove
   ① $\ker T$ is a subspace of V.
   ② $\operatorname{Im} T$ is a subspace of W.

Answers

1. $(C,+)$ is an abelian group ie complex number is closed under $+$
$(C,\cdot)$ is an abelian group.
$\forall\ a,b,c \in C$
$a\cdot(b+c) = (a\cdot b) + (a\cdot c)$
ie It is distributive

2. $S = \{(1,0,0)\ (0,1,0)\ (0,0,1)\}$
To prove that
① S should be linearly independent
② S spans R^3

$S = \{e_1, e_2, e_3\}$

$e_1 = (1,0,0)$, $e_2 = (0,1,0)$, $e_3 = (0,0,1)$

Let $(a,b,c) \in R^3$ ~~ae~~

$(a,b,c) = ae_1 + be_2 + ce_2$

$\alpha e_1 + \alpha_2 e_2 + \alpha_3 e_3 = 0 \Rightarrow (\alpha_1, \alpha_2, \alpha_3) = (0,0,0)$

$\therefore \alpha_1 = \alpha_2 = \alpha_3 = 0 \Rightarrow$ Linearly independent.

$L(S) = R^3$

Let $(a,b,c) \in R^3$

$(a,b,c) = x(1,0,0) + y(0,1,0) + z(0,0,1)$

$x = a$
$y = b$ $\therefore S$ spans $R^3 \Rightarrow$
$z = c$

3. ~~To be a vector~~ space, Z must ~~be an abelian grp w.r.to~~ $+$ ~~However~~

4. Consider $a(1,2,3) + b(3,1,2) + c(2,4,6) = (0,0,0)$

$a + 3b + 2c = 0$
$2a + b + 4c = 0$
$3a + 2b + 6c = 0$

$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 4 \\ 3 & 2 & 6 \end{bmatrix}$ $R_2 \longrightarrow R_2 - 2R_1$
$R_3 \longrightarrow R_3 - 3R_1$

$$\sim\begin{bmatrix}1 & 3 & 2\\ 0 & -5 & 0\\ 0 & -7 & 0\end{bmatrix} \quad \begin{matrix}R_2 \to R_2/-5\\ R_3 \to R_3 + 7R_2\end{matrix}$$

$$\sim\begin{bmatrix}1 & 3 & 2\\ 0 & 1 & 0\\ 0 & 0 & 0\end{bmatrix}$$

Rank = 2 < no. of unknowns.

$\Rightarrow$ System has consistent and infinite solution

$\therefore$ not linearly independent.

$Z[C]$

Let $a+ib \in C$

$x \in Z$

$x + a + ib = (a+x) + ib \in C$.

& $\notin Z$

$\therefore$ $Z(C)$ is not a vector space.

To prove $\dim(V) = 9$

We hv to show that, V has a basis consistency ~~of consists~~ of 9 elements and linearly independent.

consider $e_1 = \begin{bmatrix}1&0&0\\0&0&0\\0&0&0\end{bmatrix}$ $e_2 = \begin{bmatrix}0&1&0\\0&0&0\\0&0&0\end{bmatrix}$ $e_3 = \begin{bmatrix}0&0&1\\0&0&0\\0&0&0\end{bmatrix}$

$e_4 = \begin{bmatrix}0&0&0\\1&0&0\\0&0&0\end{bmatrix}$ $e_5 = \begin{bmatrix}0&0&0\\0&1&0\\0&0&0\end{bmatrix}$ $e_6 = \begin{bmatrix}0&0&0\\0&0&1\\0&0&0\end{bmatrix}$

$e_7 = \begin{bmatrix}0&0&0\\0&0&0\\1&0&0\end{bmatrix}$ $e_8 = \begin{bmatrix}0&0&0\\0&0&0\\0&1&0\end{bmatrix}$ $e_9 = \begin{bmatrix}0&0&0\\0&0&0\\0&0&1\end{bmatrix}$

$S = \{e_1, e_2, \ldots e_9\}$ is a basis

① Linearly independent

$$ae_1 + be_2 \cdots = \begin{bmatrix}0&0&0\\0&0&0\\0&0&0\end{bmatrix}$$

$\therefore$ $a = b = c = 0 \Rightarrow S$ is linearly independent.

② S spans V $L(S) = V$

Let $\begin{bmatrix}a&b&c\\d&e&f\\g&h&i\end{bmatrix} \in V$

then $\begin{bmatrix}a&b&c\\d&e&f\\g&h&i\end{bmatrix} = ae_1 + be_2 \cdots ie_9$

$\therefore$ $L(S) = V$

$\therefore$ $S = \{e_1, e_2 \ldots e_9\}$ is a basis of V

$\therefore$ $\dim(V) = n(S) = 9$

6. $w_1 = \dfrac{v_1}{\|v_1\|}$ $\quad v_1 = (1,1,2)$ $\quad v_2 = (2,1,3)$

$\|v_1\| = \sqrt{6}$

$= \left(\frac{1}{\sqrt6}, \frac{1}{\sqrt6}, \frac{2}{\sqrt6}\right)$

$u_2 = v_2 - \langle v_2, w_1\rangle w_1 = (2,1,3) - \left(\frac{2}{\sqrt6} + \frac{1}{\sqrt6} + \frac{6}{\sqrt6}\right)\left(\frac{1}{\sqrt6}, \frac{1}{\sqrt6}, \frac{2}{\sqrt6}\right)$

$= (2,1,3) - \left[\frac{9}{\sqrt6}\right]\left(\frac{1}{\sqrt6}, \frac{1}{\sqrt6}, \frac{2}{\sqrt6}\right)$

$= (2,1,3) - (3/2, 3/2, 3)$

$= \left(\frac{1}{2}, \frac{-1}{2}, 0\right)$

$w_2 = \frac{u_2}{\|u_2\|} = \sqrt{2}\left(\frac{\cancel{3}1}{2}, \frac{-\cancel{3}1}{2}, \cancel{3}0\right) = \cancel{\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}}, 3\sqrt{2}}\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0\right)$

7\. 1 Let $v_1, v_2 \in \ker T$

$T(v_1) = 0$ and $T(v_2) = 0$

Let $\alpha \in F$, $T(\alpha v_1) = \alpha T(v_1) = \alpha(0) = 0$

$T(v_1 + v_2) = T(v_1) + T(v_2) = 0$ since T is ~~a~~ Linear transformation

$\therefore \alpha v_1 \in \ker T$ and $v_1 + v_2 \in \ker T$

$\Rightarrow \ker T$ is a subspace of V

2\. Let $w_1, w_2 \in \operatorname{Im} T$

$\Rightarrow \exists\ v_1, v_2 \in V$ such that $T(v_1) = w_1$ & $T(v_2) = w_2$

$\alpha \in F$, $\alpha w_1 = \alpha T(v_1) = T(\alpha v_1)$ $\because T$ is LT

$\Rightarrow \alpha w_1$ is image of αv_1 under ~~LT~~ T

$\alpha w_1 \in W$

$w_1 + w_2 = T(v_1) + T(v_2) = T(v_1 + v_2)$ since T is a LT

$\Rightarrow w_1 + w_2$ is image of $v_1 + v_2$ under T

$\Rightarrow w_1 + w_2 \in W$

$\therefore \operatorname{Im} T$ is a subspace of W